

SEMESTER EXAMINATION JULY 2023
CLASS: B.Tech. SEMESTER: II
PAPER CODE: BEM-C 202
PAPER TITLE: ENGINEERING MATHEMATICS II

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 x 6 = 30 Marks)

Question-1: ✓ Solve $(1 + 3e^{x/y})dx + 3e^{x/y}(1 - \frac{x}{y})dy = 0$.

Question-2: ✓ Solve $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 2y = \cos x$.

Question-3: ✓ Solve $4\frac{\partial^2 z}{\partial x^2} + 12\frac{\partial^2 z}{\partial x \partial y} + 9\frac{\partial^2 z}{\partial y^2} = e^{3x-2y}$.

Question-4: Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = x^2 y^2$.

Question-5: Prove that $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$.

Question-6: Prove that $\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1}$, if $m = n$.

Question-7: ✓ Find the Fourier series for $f(x) = \begin{cases} -1, & -\pi < x < 0 \\ 0, & x = 0 \\ 1, & 0 < x < \pi \end{cases}$

Question-8: Find the half range sine series for $f(x) = x^2$ in the interval $0 < x < 3$.

Question-9: Find the rank correlation coefficient for the following data:

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

Question-10: Assume that on an average one telephone number out of 15 called between 2 P.M. and

3 P.M. on a week-days is busy. What is the probability that if 6 randomly selected telephone numbers are called (i) not more than 3, (ii) at least 3 of them will be busy?

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 x 10 = 40 Marks)

Question-1: Solve by changing the independent variable $x \frac{d^2y}{dx^2} + (4x^2 - 1) \frac{dy}{dx} + 4x^3y = 2x^3$.

Question-2: Solve $x^3 \frac{d^3y}{dx^3} + 3x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = x + \log x$.

Question-3: Solve by the method of separation of variable $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial y} + u$, where
 $u(x, 0) = 3e^{-5x} - 2e^{-3x}$.

Question-4: Solve in series the equation $(1 + x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 0$.

Question-5: Prove that $P_n(x)$ is the coefficient of z^n in the expansion of $(1 - 2xz + z^2)^{-1/2}$ in ascending powers of z .

Question-6: Find the Fourier series for $f(x) = x \cos x$, $-\pi < x < \pi$.

Question-7: Solve $6 \frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = 132 \log(x + 3y)$.

Question-8: In a partially destroyed laboratory record of an analysis of a correlation data, the following results only are eligible: Variance of $x = 9$, regression equations:
 $8x - 10y + 66 = 0$, $40x - 18y = 214$. What were (a) the mean values of x and y ,
(b) the standard deviation of y and (c) the coefficient of correlation between x and y .